

1. Consider the following statements :

Statement-I :

The atmosphere is heated more by incoming solar radiation than by terrestrial radiation.

Statement-II :

Carbon dioxide and other greenhouse gases in the atmosphere are good absorbers of long wave radiation.

Which one of the following is correct in respect of the above statements ?

- (a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
- (b) Both Statement-I and Statement-II are correct, but Statement-II does *not* explain Statement-I
- (c) Statement-I is correct, but Statement-II is incorrect
- (d) Statement-I is incorrect, but Statement-II is correct

2. Consider the following statements :

Statement-I :

Thickness of the troposphere at the equator is much greater as compared to poles.

Statement-II :

At the equator, heat is transported to great heights by strong convectional currents.

Which one of the following is correct in respect of the above statements ?

- (a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
- (b) Both Statement-I and Statement-II are correct, but Statement-II does *not* explain Statement-I
- (c) Statement-I is correct, but Statement-II is incorrect
- (d) Statement-I is incorrect, but Statement-II is correct

3. Consider the following :

- 1. Pyroclastic debris
- 2. Ash and dust
- 3. Nitrogen compounds
- 4. Sulphur compounds

How many of the above are products of volcanic eruptions ?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

4. Which of the following is/are correct inference/inferences from isothermal maps in the month of January ?

- 1. The isotherms deviate to the north over the ocean and to the south over the continent.
- 2. The presence of cold ocean currents, Gulf Stream and North Atlantic Drift make the North Atlantic Ocean colder and the isotherms bend towards the north.

Select the answer using the code given below :

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

5. Which of the following countries are well known as the two largest cocoa producers in the world ?

- (a) Algeria and Morocco
- (b) Botswana and Namibia
- (c) Côte d'Ivoire and Ghana
- (d) Madagascar and Mozambique

6. With reference to the Himalayan rivers joining the Ganga downstream of Prayagraj from West to East, which one of the following sequences is correct ?

- (a) Ghaghara – Gomati – Gandak – Kosi
- (b) Gomati – Ghaghara – Gandak – Kosi
- (c) Ghaghara – Gomati – Kosi – Gandak
- (d) Gomati – Ghaghara – Kosi – Gandak

7. Consider the following statements :

Statement-I :

Rainfall is one of the reasons for weathering of rocks.

Statement-II :

Rain water contains carbon dioxide in solution.

Statement-III :

Rain water contains atmospheric oxygen.

Which one of the following is correct in respect of the above statements ?

- (a) Both Statement-II and Statement-III are correct and both of them explain Statement-I
- (b) Both Statement-II and Statement-III are correct, but only one of them explains Statement-I
- (c) Only one of the Statements II and III is correct and that explains Statement-I
- (d) Neither Statement-II nor Statement-III is correct

8. Consider the following countries :

- 1. Finland
- 2. Germany
- 3. Norway
- 4. Russia

How many of the above countries have a border with the North Sea ?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

9. Consider the following information :

	<i>Waterfall</i>	<i>Region</i>	<i>River</i>
1.	Dhuandhar	Malwa	Narmada
2.	Hundru	Chota Nagpur	Subarnarekha
3.	Gersoppa	Western Ghats	Netravati

In how many of the above rows is the given information correctly matched ?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

10. Consider the following information :

	<i>Region</i>	<i>Name of the mountain range</i>	<i>Type of mountain</i>
1.	Central Asia	Vosges	Fold mountain
2.	Europe	Alps	Block mountain
3.	North America	Appalachians	Fold mountain
4.	South America	Andes	Fold mountain

In how many of the above rows is the given information correctly matched ?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

11. Consider the following airports :

1. Donyi Polo Airport
2. Kushinagar International Airport
3. Vijayawada International Airport

In the recent past, which of the above have been constructed as Greenfield projects ?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

12. With reference to "water vapour", which of the following statements is/are correct ?

1. It is a gas, the amount of which decreases with altitude.
2. Its percentage is maximum at the poles.

Select the answer using the code given below :

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

13. Consider the following description :

1. Annual and daily range of temperatures is low.
2. Precipitation occurs throughout the year.
3. Precipitation varies between 50 cm – 250 cm.

What is this type of climate ?

- (a) Equatorial climate
- (b) China type climate
- (c) Humid subtropical climate
- (d) Marine West coast climate

14. With reference to "Coriolis force", which of the following statements is/are correct ?

1. It increases with increase in wind velocity.
2. It is maximum at the poles and is absent at the equator.

Select the answer using the code given below :

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

15. On June 21 every year, which of the following latitude(s) experience(s) a sunlight of more than 12 hours ?

1. Equator
2. Tropic of Cancer
3. Tropic of Capricorn
4. Arctic Circle

Select the correct answer using the code given below :

- (a) 1 only
- (b) 2 only
- (c) 3 and 4
- (d) 2 and 4

16. One of the following regions has the world's largest tropical peatland, which holds about three years worth of global carbon emissions from fossil fuels; and the possible destruction of which can exert detrimental effect on the global climate. Which one of the following denotes that region ?

- (a) Amazon Basin
- (b) Congo Basin
- (c) Kikori Basin
- (d) Rio de la Plata Basin

17. कई उपभोक्ता उत्पादों के निर्माण के लिए प्रयुक्त होने वाले परफ्लुओरोएल्किल और पॉलिफ्लुओरोएल्किल पदार्थों (PFAS) के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए :

1. PFAS पेयजल, खाद्य और खाद्य पैकेजिंग सामग्रियों में व्यापक रूप से पाए जाते हैं ।
2. PFAS पर्यावरण में आसानी से निम्नीकृत (डिग्रेडेड) नहीं होते हैं ।
3. PFAS के लगातार संपर्क के परिणामस्वरूप जंतुओं के शरीर में जैवसंचय हो सकता है ।

उपर्युक्त कथनों में से कौन-से सही हैं ?

- (a) केवल 1 और 2
- (b) केवल 2 और 3
- (c) केवल 1 और 3
- (d) 1, 2 और 3

18. निम्नलिखित पर विचार कीजिए :

1. कैराबिड बीटल्स
2. कांतर (सेन्टिपीड्स)
3. मक्खियाँ
4. दीमक
5. बर् (वास्प)

उपर्युक्त जीवों के कितने प्रकार में परजीव्याभ जातियाँ (पैरासीटॉइड स्पीशीज़) पाई जाती हैं ?

- (a) केवल दो
- (b) केवल तीन
- (c) केवल चार
- (d) सभी पाँच

19. निम्नलिखित पौधों पर विचार कीजिए :

1. मूँगफली
2. कुलथी (हॉर्स-ग्राम)
3. सोयाबीन

उपर्युक्त में से कितने मटर कुल (फैमिली) के हैं ?

- (a) केवल एक
- (b) केवल दो
- (c) सभी तीन
- (d) कोई नहीं

20. निम्नलिखित कथनों पर विचार कीजिए :

कथन-I :

इंडियन फ्लाइंग फॉक्स को वन्य जीव (संरक्षण) अधिनियम, 1972 के अधीन “पीड़क जंतु” की श्रेणी में रखा गया है ।

कथन-II :

इंडियन फ्लाइंग फॉक्स अन्य जंतुओं का रक्त पीता है ।

उपर्युक्त कथनों के संबंध में, निम्नलिखित में से कौन-सा सही है ?

- (a) कथन-I और कथन-II दोनों सही हैं तथा कथन-II, कथन-I की व्याख्या करता है
- (b) कथन-I और कथन-II दोनों सही हैं, किंतु कथन-II, कथन-I की व्याख्या नहीं करता है
- (c) कथन-I सही है, किंतु कथन-II सही नहीं है
- (d) कथन-I सही नहीं है, किंतु कथन-II सही है

17. With reference to perfluoroalkyl and polyfluoroalkyl substances (PFAS) that are used in making many consumer products, consider the following statements :

1. PFAS are found to be widespread in drinking water, food and food packaging materials.
2. PFAS are not easily degraded in the environment.
3. Persistent exposure to PFAS can lead to bioaccumulation in animal bodies.

Which of the statements given above are correct ?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

18. Consider the following :

1. Carabid beetles
2. Centipedes
3. Flies
4. Termites
5. Wasps

Parasitoid species are found in how many of the above kind of organisms ?

- (a) Only two
- (b) Only three
- (c) Only four
- (d) All five

19. Consider the following plants :

1. Groundnut
2. Horse-gram
3. Soybean

How many of the above belong to the pea family ?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

20. Consider the following statements :

Statement-I :

The Indian Flying Fox is placed under the "vermin" category in the Wild Life (Protection) Act, 1972.

Statement-II :

The Indian Flying Fox feeds on the blood of other animals.

Which one of the following is correct in respect of the above statements ?

- (a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
- (b) Both Statement-I and Statement-II are correct, but Statement-II does **not** explain Statement-I
- (c) Statement-I is correct, but Statement-II is incorrect
- (d) Statement-I is incorrect, but Statement-II is correct

21. The organisms "Cicada, Froghopper and Pond skater" are :

- (a) Birds
- (b) Fish
- (c) Insects
- (d) Reptiles

22. Consider the following statements :

Statement-I :

Many chewing gums found in the market are considered a source of environmental pollution.

Statement-II :

Many chewing gums contain plastic as gum base.

Which one of the following is correct in respect of the above statements ?

- (a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
- (b) Both Statement-I and Statement-II are correct, but Statement-II does **not** explain Statement-I
- (c) Statement-I is correct, but Statement-II is incorrect
- (d) Statement-I is incorrect, but Statement-II is correct

23. Consider the following pairs :

	Country	Animal found in its natural habitat
1.	Brazil	Indri
2.	Indonesia	Elk
3.	Madagascar	Bonobo

How many of the pairs given above are correctly matched ?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

24. Consider the following statements regarding World Toilet Organization :

- 1. It is one of the agencies of the United Nations.
- 2. World Toilet Summit, World Toilet Day and World Toilet College are the initiatives of this organization, to inspire action to tackle the global sanitation crisis.
- 3. The main focus of its function is to grant funds to the least developed countries and developing countries to achieve the end of open defecation.

Which of the statements given above is/are correct ?

- (a) 2 only
- (b) 3 only
- (c) 1 and 2
- (d) 2 and 3

25. Consider the following statements :

- 1. Lions do not have a particular breeding season.
- 2. Unlike most other big cats, cheetahs do not roar.
- 3. Unlike male lions, male leopards do not proclaim their territory by scent marking.

Which of the statements given above are correct ?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

26. Which one of the following is the correct description of "100 Million Farmers" ?

- (a) It is a platform for accelerating the transition towards food and water systems that are net-zero (carbon), nature-positive and that aims to increase farmer resilience.
- (b) It is an international alliance and a network of individuals and farming organisations interested in supporting and strengthening the development of the organic animal husbandry.
- (c) It is a digital platform fully integrated with service providers and built on blockchain that lets buyers, sellers and third parties trade fertilizers quickly and securely.
- (d) It is a platform with the mission of encouraging the farmers to form Farmer Product Organisations or Agribusiness Consortia, thus facilitating the access to global open markets to sell their products.

27. Consider the following :

- 1. Battery storage
- 2. Biomass generators
- 3. Fuel cells
- 4. Rooftop solar photovoltaic units

How many of the above are considered "Distributed Energy Resources" ?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

28. Which one of the following shows a unique relationship with an insect that has coevolved with it and that is the only insect that can pollinate this tree ?

- (a) Fig
- (b) Mahua
- (c) Sandalwood
- (d) Silk cotton

29. Consider the following :

- 1. Butterflies
- 2. Fish
- 3. Frogs

How many of the above have poisonous species among them ?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

30. Consider the following :

- 1. Cashew
- 2. Papaya
- 3. Red sanders

How many of the above trees are actually native to India ?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

31. With reference to radioisotope thermoelectric generators (RTGs), consider the following statements :

1. RTGs are miniature fission reactors.
2. RTGs are used for powering the onboard systems of spacecrafts.
3. RTGs can use Plutonium-238, which is a by-product of weapons development.

Which of the statements given above are correct ?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

32. Consider the following statements :

Statement-I :

Giant stars live much longer than dwarf stars.

Statement-II :

Compared to dwarf stars, giant stars have a greater rate of nuclear reactions.

Which one of the following is correct in respect of the above statements ?

- (a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
- (b) Both Statement-I and Statement-II are correct, but Statement-II does **not** explain Statement-I
- (c) Statement-I is correct, but Statement-II is incorrect
- (d) Statement-I is incorrect, but Statement-II is correct

33. Which one of the following is synthesised in human body that dilates blood vessels and increases blood flow ?

- (a) Nitric oxide
- (b) Nitrous oxide
- (c) Nitrogen dioxide
- (d) Nitrogen pentoxide

34. Consider the following activities :

1. Identification of narcotics on passengers at airports or in aircraft
2. Monitoring of precipitation
3. Tracking the migration of animals

In how many of the above activities can the radars be used ?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

35. Consider the following aircraft :

1. Rafael
2. MiG-29
3. Tejas MK-1

How many of the above are considered fifth generation fighter aircraft ?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

36. In which of the following are hydrogels used ?
1. Controlled drug delivery in patients
 2. Mobile air-conditioning systems
 3. Preparation of industrial lubricants
- Select the correct answer using the code given below :
- (a) 1 only
 - (b) 1 and 2 only
 - (c) 2 and 3 only
 - (d) 1, 2 and 3
37. Which one of the following is the exhaust pipe emission from Fuel Cell Electric Vehicles, powered by hydrogen ?
- (a) Hydrogen peroxide
 - (b) Hydronium
 - (c) Oxygen
 - (d) Water vapour
38. Recently, the term “pumped-storage hydropower” is actually and appropriately discussed in the context of which one of the following ?
- (a) Irrigation of terraced crop fields
 - (b) Lift irrigation of cereal crops
 - (c) Long duration energy storage
 - (d) Rainwater harvesting system
39. “Membrane Bioreactors” are often discussed in the context of :
- (a) Assisted reproductive technologies
 - (b) Drug delivery nanotechnologies
 - (c) Vaccine production technologies
 - (d) Wastewater treatment technologies
40. With reference to the Indian economy, “Collateral Borrowing and Lending Obligations” are the instruments of :
- (a) Bond market
 - (b) Forex market
 - (c) Money market
 - (d) Stock market

41. The total fertility rate in an economy is defined as :

- (a) the number of children born per 1000 people in the population in a year.
- (b) the number of children born to a couple in their lifetime in a given population.
- (c) the birth rate minus death rate.
- (d) the average number of live births a woman would have by the end of her child-bearing age.

42. Consider the following statements :

1. In India, Non-Banking Financial Companies can access the Liquidity Adjustment Facility window of the Reserve Bank of India.
2. In India, Foreign Institutional Investors can hold the Government Securities (G-Secs).
3. In India, Stock Exchanges can offer separate trading platforms for debts.

Which of the statements given above is/are correct ?

- (a) 1 and 2 only
- (b) 3 only
- (c) 1, 2 and 3
- (d) 2 and 3 only

43. In India, which of the following can trade in Corporate Bonds and Government Securities ?

1. Insurance Companies
2. Pension Funds
3. Retail Investors

Select the correct answer using the code given below :

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

44. Consider the following :

1. Exchange-Traded Funds (ETF)
2. Motor vehicles
3. Currency swap

Which of the above is/are considered financial instruments ?

- (a) 1 only
- (b) 2 and 3 only
- (c) 1, 2 and 3
- (d) 1 and 3 only

45. With reference to the sectors of the Indian economy, consider the following pairs :

	<i>Economic activity</i>	<i>Sector</i>
1.	Storage of agricultural produce	Secondary
2.	Dairy farm	Primary
3.	Mineral exploration	Tertiary
4.	Weaving cloth	Secondary

How many of the pairs given above are correctly matched ?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

46. Consider the following materials :

1. Agricultural residues
2. Corn grain
3. Wastewater treatment sludge
4. Wood mill waste

Which of the above can be used as feedstock for producing Sustainable Aviation Fuel ?

- (a) 1 and 2 only
- (b) 3 and 4 only
- (c) 1, 2, 3 and 4
- (d) 1, 3 and 4 only

47. With reference to physical capital in Indian economy, consider the following pairs :

	Items	Category
1.	Farmer's plough	Working capital
2.	Computer	Fixed capital
3.	Yarn used by the weaver	Fixed capital
4.	Petrol	Working capital

How many of the above pairs are correctly matched ?

48. Which one of the following words/phrases is most appropriately used to denote "an interoperable network of 3D virtual worlds that can be accessed simultaneously by millions of users, who can exert property rights over virtual items" ?

- (a) Big data analytics
- (b) Cryptography
- (c) Metaverse
- (d) Virtual matrix

49. With reference to the rule/rules imposed by the Reserve Bank of India while treating foreign banks, consider the following statements :

- 1. There is no minimum capital requirement for wholly owned banking subsidiaries in India.
- 2. For wholly owned banking subsidiaries in India, at least 50% of the board members should be Indian nationals.

Which of the statements given above is/are correct ?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

50. With reference to Corporate Social Responsibility (CSR) rules in India, consider the following statements :

- 1. CSR rules specify that expenditures that benefit the company directly or its employees will not be considered as CSR activities.
- 2. CSR rules do not specify minimum spending on CSR activities.

Which of the statements given above is/are correct ?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2